

Future Mountain Rockynook (SBFR) Fire Output Metadata

Preliminary fire spread output for Rockynook (a sub-watershed of the Santa Barbara Front Range)

Simulations were run with WRF climate data

Individual Cube Information

Cube	Patch Family (zoneID)	PatchID	Reference	% of Patch Family	Overstory	Understory
North	14094	1409401	patch 1	50%	Chaparral	Grass
North	14094	1409402	patch 2	50%	Chaparral	Grass
South	13260	1326001	patch 1	15%	Chaparral	Grass
South	13260	1326002	patch 2	85%	Grass	NA
High	6487	648701	patch 1	50%	Chaparral	Grass
High	6487	648702	patch 2	50%	Chaparral	Grass
Low	18030	1803001	patch 1	50%	Chaparral	Grass
Low	18030	1803002	patch 2	50%	Chaparral	Grass
Riparian	20803	2080301	patch 1	50%	Oak	Grass
Riparian	20803	2080301	patch 2	50%	Chaparral	Grass

File Naming conventions:

p = patch
b = basin
m = monthly
d = daily
1 = patch 1
2 = patch 2

i.e. cubes_p2d: the p2d in the name = patch 2 daily output (daily patch output for patch 2 of the patch family)

File Name	Spatial level	Temporal level	Description
allPatches_p1m	Patch	Monthly	output for All patches in the watershed; output for patch 1 in each Patch Family (ZoneID)

File Name	Spatial level	Temporal level	Description
allPatches_p2m	Patch	Monthly	output for All patches in the watershed; output for patch 2 in each Patch Family (ZoneID)
aggregate_cube_bd	Basin	Daily	Average Basin daily output for the aggregate cube
cubes_p1d	Patch	Daily	Individual cube Patch/Stratum output for patch 1 in the Patch Family (ZoneID)
cubes_p2d	Patch	Daily	Individual cube Patch/Stratum output for patch 2 in the Patch Family (ZoneID)

Variables

File Names (prefix)	Spatial level	Temporal level	Variable	Description
allPatches, cubes	patch	monthly, daily	zoneID	Zone identifier - this is also equivalent to the Patch Family identifier
allPatches, cubes	patch	monthly, daily	patchID	Patch identifier indicating which patch in the family (patch 1 ends in 01 - patch 2 ends in 02)
allPatches_pXm	patch	monthly	totalCover	Overstory total carbon including live/dead vegetation components & non-living organic matter such as litter and woody debris
allPatches_pXm	patch	monthly	plantCover	Overstory total plant carbon
allPatches_pXm	patch	monthly	totalCunder	Understory total carbon (live/dead vegetation, litter, and woody debris)
allPatches_pXm	patch	monthly	plantCunder	Understory total plant carbon
allPatches, cubes	patch	monthly, daily	burned	TBD - this is either the % that burned, or psread (Probability of fire spread to a neighboring patch)
allPatches, cubes	patch	monthly, daily	fire	1 = patch experienced fire, 0 = no fire occurred
cubes_pXd	patch	daily	coverfract	Percent a patch occupies of the patch family
aggregate, cubes	basin, patch	daily	litterC	Total above ground Litter Carbon

File Names (prefix)	Spatial level	Temporal level	Variable	Description
aggregate, cubes	basin, patch	daily	soilC	Total Soil Carbon
aggregate, cubes	basin, patch	daily	depthToGW	The depth to the water table below the ground surface
aggregate, cubes	basin, patch	daily	canopyevap	Evaporation from the vegetation canopy
aggregate, cubes	basin, patch	daily	groundevap	Evaporation from the soil surface
aggregate_cube_b	basin	daily	streamflow	Amount of basin water discharge
aggregate, cubes	basin, patch	daily	vegAccessWater	Vegetation access to water - the amount of water currently stored within the root zone that is available for plant uptake
cubes_pXd	patch	daily	Qin	Lateral subsurface water entering patch from upslope patches
cubes_pXd	patch	daily	Qout	Lateral subsurface water leaving the patch to downslope neighbors
aggregate_cube_b	basin	daily	netpsn	Net photosynthesis of all vegetation
cubes_pXd	patch	daily	netpsnOver	Net photosynthesis of overstory vegetation
cubes_pXd	patch	daily	netpsnUnder	Net photosynthesis of understory vegetation
aggregate_cube_b	basin	daily	GPP	Gross photosynthesis of all vegetation
cubes_pXd	patch	daily	GPP_over	Gross photosynthesis of overstory vegetation
cubes_pXd	patch	daily	GPP_under	Gross photosynthesis of understory vegetation
aggregate_cube_b	basin	daily	respiration	Total respiration of all vegetation
cubes_pXd	patch	daily	respOver	Overstory plant respiration
cubes_pXd	patch	daily	respUnder	Understory plant respiration
aggregate_cube_b	basin	daily	height	Average height of all vegetation
cubes_pXd	patch	daily	heightOver	Overstory vegetation height
cubes_pXd	patch	daily	heightUnder	Understory vegetation height
aggregate_cube_b	basin	daily	transpiration	Total amount of water transpired by all vegetation

File Names (prefix)	Spatial level	Temporal level	Variable	Description
cubes_pXd	patch	daily	transOver	Amount of water transpired by overstory vegetation
cubes_pXd	patch	daily	transUnder	Amount of water transpired by understory vegetation
aggregate_cube_b	basin	daily	leafC	Total leaf carbon for all vegetation
cubes_pXd	patch	daily	leafC_over	Overstory leaf carbon
cubes_pXd	patch	daily	leafC_under	Understory leaf carbon
aggregate_cube_b	basin	daily	stemC	Total stem carbon for all vegetation
cubes_pXd	patch	daily	stemC_over	Overstory stem carbon
aggregate_cube_b	basin	daily	rootC	Total root carbon for all vegetation
cubes_pXd	patch	daily	rootC_over	Overstory root carbon
cubes_pXd	patch	daily	rootC_under	Understory root carbon
aggregate_cube_b	basin	daily	rootdepth	Average root zone depth (depth of the effective root zone accessible to plant roots for water uptake)
cubes_pXd	patch	daily	rootdepthOver	Overstory root zone depth
cubes_pXd	patch	daily	rootdepthUnder	Understory root zone depth
cubes_pXd	patch	daily	laiOver	Overstory leaf area index
cubes_pXd	patch	daily	laiUnder	Understory leaf area index
cubes_pXd	patch	daily	rain	patch precipitation received
cubes_pXd	patch	daily	tmax	patch maximum temperature
cubes_pXd	patch	daily	tmin	patch minimum temperature
cubes_pXd	patch	daily	wind speed	wind speed
cubes_pXd	patch	daily	wind direction	patch wind direction
cubes_pXd	patch	daily	RH	patch relative humidity